
Counterdiabatic Hamiltonian Monte Carlo

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Hamiltonian Monte Carlo (HMC) is a state of the art method for sampling from
2 distributions with differentiable densities, but struggles to converge to the stationary
3 distribution for challenging multimodal problems. Running HMC with a time
4 varying Hamiltonian, to interpolate from a tractable distribution to the target
5 of interest results in bias, which can be corrected by weighting schemes, as in
6 Sequential Monte Carlo sampling [11]. However, this can be inefficient, since it
7 requires slow change between the two distributions. Inspired by [26], where a
8 learned *counterdiabatic* term allows for efficient quantum state preparation, we
9 propose *Counterdiabatic Hamiltonian Monte Carlo*. We establish its relationship
10 to recent proposals for accelerating gradient-based sampling with learned drift
11 terms, and show initial promising results.

12 1 Introduction

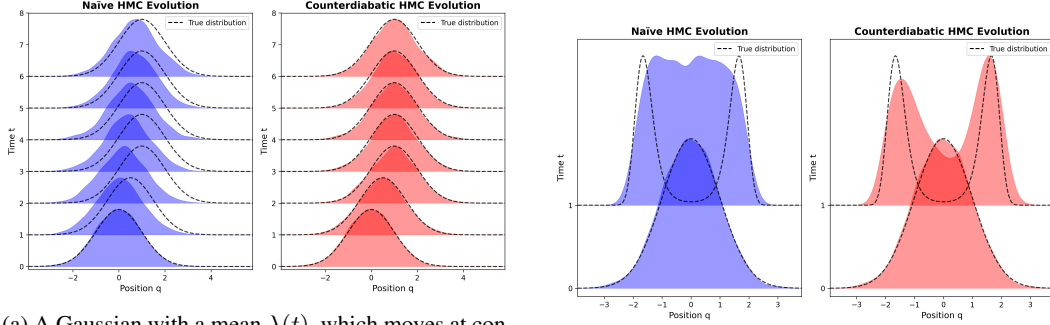
13 Algorithms for sampling from probability distributions have many application in the physical sciences
14 [30, 33] and Bayesian statistics [14]. For arbitrary distributions with differentiable density functions,
15 *Hamiltonian Monte Carlo* (HMC) [5] is a state of the art method for sampling, and in conjunction
16 with the No-U-Turn scheme for trajectory length [19], it is a reliable and widely used method [15].

17 HMC is founded on a correspondence between probabilistic inference and classical statistical physics,
18 and produces a Markov kernel based on a numerical simulation of Hamiltonian dynamics. This entails
19 an interpretation of the stationary distribution (in the extended space of position and momentum) as
20 the canonical equilibrium distribution of statistical mechanics, and interpretations of the logarithm of
21 the density to the potential energy, force to the gradient of this logarithm, and reversibility to work
22 [27].

23 MCMC works by constructing a Markov kernel which has as its stationary distribution the target
24 distribution, so that a Markov chain samples from the target distribution in the limit of many steps.
25 However, for certain distributions, the number of steps of the chain required to reach the target
26 distribution is prohibitively large. A motivating example is a mixture of two Gaussians with means
27 separated by a distance much larger than the respective variances. If the chain is initialized near one
28 of the two modes, attaining stationarity requires moving through areas of low probability density.

29 One proposed solution to this problem is a Sequential Monte Carlo (SMC) sampler [9], which
30 constructs a series of distributions, starting with an easy distribution (such as a Gaussian) and ending
31 with the target (such as the Gaussian mixture), using a Markov kernel at each step in the series and a
32 weighting scheme to remove bias. The challenge of SMC is to make each distribution in the series
33 similar enough to the previous that the output of the Markov kernel does not lag far behind the current
34 distribution (which results in a high variance weighting scheme), while making them different enough
35 that only a small number of steps is required.

36 In the setting of HMC, moving through a series of distributions in this way corresponds to time evolu-
37 tion of a physical system while changing the system’s parameters, a central topic of thermodynamics



(a) A Gaussian with a mean $\lambda(t)$, which moves at constant velocity from $\lambda = 0$ to $\lambda = 1$, and then remains at 1. Simply running HMC at each intermediate time step (blue) results in a lag, which is accounted for by the counterdiabatic driving term (red).

(b) An interpolation between a Gaussian at $t = 1$ and a double well at $t = 1$. The counterdiabatic driving does not result in a perfect transport, but clearly accelerates the process.

Figure 1: Illustration of counterdiabatic driving in Hamiltonian Monte Carlo. Left: Moving mean example showing lag and correction. Right: Double well example showing accelerated transport.

and statistical mechanics. From this perspective, varying the parameters too quickly results in *lag*, or more physically *dissipation* [31], where the Hamiltonian flow cannot return to equilibrium quickly enough as the system's parameters change.

Several papers [10, 4], both in the quantum and classical setting, have noted that one can compensate for this lag by adding a *counterdiabatic driving term* to the Hamiltonian. More recently, a scheme to approximate this term variationally has been proposed in a quantum setting [26]. We adapt this approach to a classical setting, to obtain a method for transitioning efficiently between a simple tractable distribution and a challenging distribution of interest. We show that our approach is closely related to the proposals of [1, 32, 2] among others, but in a Hamiltonian setting.

2 Related work

Sequential Monte Carlo In the context of statistics and Bayesian inference, several algorithms exist to deal with the problem of slow convergence to the target distribution. Most relevant to the present work is the application of Sequential Monte Carlo [11] to sampling from static distributions [9], by constructing a series of distributions with densities π_i where π_0 is the initial distribution and π_T is the target distribution. Here a series of particles are evolved according to a Markov kernel targeting the stationary distribution of π_i at step i , and at each step reweighting and resampling, so that the weighted particles are distributed according to p_i .

Several proposals have been made to replace the Markov kernel with one which more effectively transports the particles from π_i to π_{i+1} , including Controlled Sequential Monte Carlo [18] and Schrodinger Bridge SMC [3].

Dynamical transport A separate body of research approaches the problem from the perspective of stochastic optimal control [13] and optimal transport [25]. Here, one considers an ODE or SDE governing a variable of interest x , which gives rise to a PDE that governs the evolution of the corresponding density π . The goal is to design the ODE or SDE so that the density π evolves from a simple initial distribution to a chosen target distribution.

Among these, Controlled Monte Carlo Diffusions (CMCD [32]), the Liouville Flow Importance Sampler (LFIS [29]), Path Guided Particle Sampling (PGPS [12]), the Path-Integral Sampler (PIS [34]) and Non-equilibrium transport sampling (NETS [1]) all learn a drift term of a differential equation, parametrized by a neural network.

At the intersection of these two perspectives, several recent approaches [1, 32] have proposed to use weighting schemes to correct for remaining bias, in the manner of SMC. These are inspired by Jarsynski's equality [20] and more generally from the detailed fluctuation theorem [7].

Statistical physics Finally, the problem of transitioning from one stationary distribution to another is a core topic of thermodynamics, and has practical implications for the preparation of physical systems. Adding a term to Hamiltonian dynamics to preserve stationarity through a series of intermediate distributions has been proposed several times in a quantum setting, as *engineered swift equilibration* [22] and counterdiabatic driving [10], where the corresponding problem with a quickly changing Hamiltonian is failure to preserve eigenstates. The core ideas carry over to a classical setting [24], including with stochastic dynamics, often under the name *shortcuts to adiabaticity*, as well as *escorted free energy* (EFE) [31]. We recommend [17] for an overview. While the problem is closely related to the optimal transport and control perspectives above, the setting of Hamiltonian dynamics provides useful geometric insights, in particular that the learned correction to the Hamiltonian can be understood geometrically as a “gauge potential”, or equivalently as a connection on a fiber bundle [21].

The above papers focus on analytic solutions to the driving term for particular physical systems; meanwhile, [26] proposes *learning* a driving term in a quantum context, but notes that the approach is transferable to a classical setting, which is a central inspiration for our approach.

Contributions Since Hamiltonian Monte Carlo is the gold standard approach for sampling from differentiable densities, it is natural that schemes for accelerating convergence to stationarity should focus on it. We emphasize this focus as the main ingredient by which our method differs from the recent approaches to sampling with learning discussed above.

Table 2 summarizes what we consider to be the most similar approaches to our own and the relevant differences. All of the listed methods take a population of particles, and evolve them, while also varying the distribution from an easy initial distribution to a target of interest. *Dynamics* refers to the way in which particles are updated at each step, *Weights* to whether a weighting scheme is used to correct the distribution of particles, *Learned term* to what, if any, term needs to be learned for the dynamics, and *Amortized* to whether the algorithm requires a phase before runtime in which training takes place (as opposed to any training happening during the evolution of the particles).

Method	Dynamics	Weights	Learned term	Amortized	Field
CSMC	Markov kernel	Yes	Kernel	Yes	Stats
EFE [31]	Hamiltonian	No	None	No	Phys/Chem
APTFC [10]	Hamiltonian	No	None	No	Chem
AIS [23]	MCMC	Yes	None	No	Stats
NETS [1]	Overdamped Langevin	Yes	Drift	Yes	ML
CMCD [32]	Langevin	No (?)	Drift	Yes	ML
AFT [2]	NF + MCMC	Yes	NF Map	No	ML
LFIS [29]	ODE	No	Drift	No	ML
PGPS [12]	ODE	No	Drift	No	ML
DMT [28]	PDE	No	Drift	No	ML/PDEs
PIS [34]	SDE	Yes	Drift	Yes	ML/Control
CMC (ours)	Hamiltonian	No	Gauge field	No	ML/Phys

3 Counterdiabatic Monte Carlo

Hamiltonian dynamics Suppose we have a physical system given by a d dimensional vector $q \in \mathbb{R}^d$, such as the position of a particle. Hamilton’s equations of motion are a 1st order ODE defined on $(q, p) \in \mathbb{R}^{2d}$, where p is a second variable called the momentum. The ODE is defined in terms of a real-valued *Hamiltonian* function $H(q, p)$.

$$\dot{q} = \nabla_p H(q, p) \quad (1)$$

$$\dot{p} = -\nabla_q H(q, p) \quad (2)$$

We refer to the solution to the ODE as the *flow* ϕ_H , so that $\phi_H(q_0, p_0, t)$ is the state of the system at time t given the initial state (q_0, p_0) .

Poisson bracket If f is also a real valued function of (q, p) , then the time derivative $\partial_t f$ can be written as the *Poisson bracket* $\partial_t f(q, p) = \{f, H\}(q, p) := \nabla_q f \cdot \nabla_p H - \nabla_p f \cdot \nabla_q H$.

We note that the Poisson bracket is antisymmetric, i.e. $\{A, B\} = -\{B, A\}$, and that $\int C(q, p)\{A, B\}(q, p)dqdp = \int B(q, p)\{C, A\}(q, p)dqdp$.

Canonical distribution The *canonical distribution* defined as $\rho_H(q, p) \propto e^{-H(q, p)}$ is a stationary distribution of Hamiltonian dynamics. Indeed, the continuity equation $\partial_t \rho_H = \{H, \rho_H\}$ implies that $\partial_t \rho_H = 0$, by an above property of the Poisson bracket. We denote by ϕ_H^* the flow of the differential equation, so that $\phi_H^*(t')(\rho_H) = \rho_H$.

Hamiltonian Monte Carlo The principle of Hamiltonian Monte Carlo is to use a discretization of Hamiltonian dynamics to produce a kernel which has a stationary distribution ρ_H for which the marginal on q , i.e. $\pi(q) \propto \int \rho_H(q, p)dp$, is the distribution of interest. This can be achieved by choosing $H = -\nabla \log \pi(q) + \frac{1}{2}|p|^2$, which only requires knowing the density up to the normalization constant.

The Metropolis-Hastings algorithm [6] is used to correct for bias coming from the discretization of the Hamiltonian dynamics, and momentum can be refreshed every n steps, or partially refreshed every step, in order to ensure ergodicity.

3.1 Counterdiabatic driving

When H is time dependent, for example through a parameter λ , we can consider the time dependent flow $\phi_{H(\lambda(t))}$. Crucially, we note that $\partial_t \rho_{H(\lambda(t))} \neq 0$, so that $\phi_H^*(t')\rho_{H(\lambda(t))} \neq \rho_{H(\lambda(t+t'))}$. In other words, the flow of the time-varying Hamiltonian does not preserve the time-varying distribution $\rho_{H(\lambda(t))}$. If it did, we could simply run Hamiltonian dynamics with $H(\lambda(t))$ and obtain at time T the intended target perfectly. To see why it fails, we proceed as follows:

$$\begin{aligned} \partial_t \rho_{H(\lambda(t))} &= \{H, \rho_{H(\lambda(t))}\} + \dot{\lambda} \partial_\lambda \rho_{H(\lambda(t))} \\ &= 0 - \dot{\lambda} \partial_\lambda H \rho_{H(\lambda(t))} - \dot{\lambda} \frac{\partial \log Z}{\partial \lambda} \rho_{H(\lambda(t))} = (-\partial_\lambda H + \langle \partial_\lambda H \rangle_t) \dot{\lambda} \rho_{H(\lambda(t))} \end{aligned}$$

[10], [4] and [31, 16] note¹ that one can offset the lag by introducing a term A , known as the *counterdiabatic gauge potential*, with the property that $\{A, H\} = \partial_\lambda H - \langle \partial_\lambda H \rangle_t$. Then we would have:

$$\partial_\lambda \rho_{H(\lambda(t))} = \{H, A\} \rho_{H(\lambda(t))} = \{\rho_{H(\lambda(t))}, -A\} = \{A, \rho_{H(\lambda(t))}\}$$

This can be interpreted as the evolution of the distribution that would be generated by a Hamiltonian A . This demonstrates a remarkable result: if we choose as our Hamiltonian $H + \dot{\lambda}A$, we have that $\phi_{H+\dot{\lambda}A}^*(t')\rho_{H(\lambda(t))} = \rho_{H(\lambda(t+t'))}$.

Learning the counterdiabatic gauge potential [26] demonstrates that one can variationally approximate A in a computationally practical way. In classical terms, the key idea is that we minimize $\mathbb{E}_{z \sim \rho_{H(\lambda(t))}}[|G(z)|^2]$ at each time point t , where

$$G(z) = \{A, H\}(z) - \partial_\lambda H(z) + \mathbb{E}[\partial_\lambda H(z)]$$

Multiplying out, we find $\mathbb{E}[|G(z)|^2] = \mathbb{E}[\{A, H\}^2 + (\partial_\lambda H)^2 + \mathbb{E}[\partial_\lambda H]^2 - 2\{A, H\}(\partial_\lambda H) + 2\{A, H\}\mathbb{E}[\partial_\lambda H] - 2(\partial_\lambda H)\mathbb{E}[\partial_\lambda H]] = \mathbb{E}[\{A, H\}^2 + (\partial_\lambda H)^2 - 2\{A, H\}(\partial_\lambda H) + 2\{A, H\}\mathbb{E}[\partial_\lambda H] - \mathbb{E}[\partial_\lambda H]^2]$

But we can see that a minimum of the following:

$$R(z) = \langle |\{A(z), H(z)\} - \partial_\lambda H(z)|^2 \rangle = \langle \{A(z), H(z)\}^2 - 2\{A(z), H(z)\}\partial_\lambda H(z) + (\partial_\lambda H(z))^2 \rangle$$

¹Here, the focus is on quantum systems, but by the correspondence between Poisson brackets and commutators, the central insight may be translated to a classical setting, as explained in [].

140 is a minimum of $|G|^2$, since when $\partial_\lambda H = \{A, H\}$, we find that $\langle \partial_\lambda H \rangle = \langle \{A, H\} \rangle =$
 141 $\int \rho \{A, H\} dz = \int A \{\rho, H\} = 0$, where the penultimate step follows by an above property of
 142 the Poisson bracket.

143 This means that we can minimize $\mathbb{E}[R(A, H)] = \mathbb{E}[|\{A, H\} - \partial_\lambda H|^2]$ instead, which is simple to
 144 calculate.

145 3.2 Weighting

146 When performing the algorithm described above, bias, in the sense of the population of particles at
 147 time t failing to agree with $\rho_{H(t)}$ arises from two sources. The first is the numerical discretization
 148 of Hamilton's equations of motion. The second is the fact that we only approximate the true
 149 counterdiabatic driving term A .

150 Bias in of itself is not a problem; once we have approached close to the final stationary distribution,
 151 HMC with Metropolis adjustment can be used to remove remaining bias. However, since the learning
 152 of A requires high quality samples from each successive distribution, bias removal is important.

153 and emphasize that we resample using the weights along the path of distributions, rather than only
 154 using the weights This is important, since the weighted particles are used to calculate the expectation
 155 required for the learning of A . We note that this is an important difference (CHECK) to NETS.

156 Here, we follow SMC:

157 that the weight is determined by a ratio (or more technically the Radon-Nikodym derivative) of the
 158 form:

$$\frac{d(L_{t-1} \otimes \gamma_t)}{d(\gamma_{t-1} \otimes M_t)}(z_{t-1}, z_t)$$

159 In the simplest case, M can be chosen as any invariant kernel, and L as its time reversal, i.e.
 160 $L_{t-1}(z_t, z_{t-1})\pi_t(z_t) = \pi_t z_{t-1} M_t(z_{t-1}, z_t)$.

161 In many cases, the ratio of kernels derived from forward and backward dynamics is known as a result
 162 of the Crooks detailed fluctuation theorem [7]. This states that

163 Weighting schemes can remove bias, and [35] gives the general form of the weight in the presence of
 164 a counterdiabatic driving term.

$$dW = \frac{\partial H}{\partial t} - \nabla H \cdot \nabla A + \nabla^2 A$$

165 which defines $W[z(t)] = \int_0^T dW$

166 This allows us to calculate the reversibility, since [35] show that

167 4 Algorithm design

168 The fact that we can learn an approximation of A suggests an algorithm for sampling. We first
 169 choose a path of densities $\pi(q; \lambda(t))$ where $\pi(q; \lambda(0))$ is chosen to be straightforward to sample
 170 from, and where $\pi(q; \lambda(T))$ is the distribution of interest. This defines a time varying Hamiltonian
 171 $H(p, q; \lambda(t)) = -\nabla \log \pi(q; \lambda(t)) + \frac{1}{2}|p|^2$.

172 We then initialize a population of particles in $\pi(\cdot; \lambda(0))$, distributed according to $\rho_{H(\lambda(0))}$. At each
 173 time t , discretized in steps of length ϵ , we minimize the *expectation* of R , under the empirical
 174 distribution of the current population, to obtain an estimate of A (Algorithm 1). We use $H(\cdot; \lambda(t)) +$
 175 $\dot{\lambda}A(\cdot; \lambda(t))$ as our Hamiltonian at a given time t . Further, we have the option of computing weights
 176 according to (??) after each time step, which correct for the bias introduced by the numerical
 177 discretization of the dynamics and the approximation of A . This is summarized in Algorithm 2.

178 This can be understood as a special case of the general framework of Sequential Monte Carlo samplers
 179 [doucet2001sequential], as shown in appendix TODO.

Algorithm 1 Fit adiabatic gauge potential A_ϕ at a given λ

Require: Real-valued function $A_\phi(q, p; \lambda)$, optimizer (e.g. Adam), schedule λ_i , population of particles $\{q_i, p_i\}_{i=1}^N$, number of iterations T .

- 1: **for** $t = 1, \dots, T$ **do**
 - 2: Compute $l \leftarrow \frac{1}{N} \sum_{i=1}^N \{A_\phi, H_\lambda\}(q_i, p_i) - \partial_\lambda U(q_i; \lambda)$
 - 3: Update ϕ using optimizer with loss l
 - 4: **end for**
 - 5: **return** ϕ
-

Algorithm 2 Counterdiabatic HMC (CHMC) with population and weighting

Require: Population $\{q_i^{(0)}\}_{i=1}^N$ from initial distribution; potential $U(q; \lambda)$ and gradients; step size ε ; schedule $\{\lambda_k\}_{k=1}^L$ with $\lambda_1 = 0$ (easy) and $\lambda_L = 1$ (target); learned $A_\phi(q, p; \lambda)$ and its partials $\nabla_q A_\phi, \nabla_p A_\phi$.

- 1: Initialize weights $w_i^{(0)} = 1/N$ for all i
- 2: **for** $k = 1$ to $L - 1$ **do**
- 3: **for** $i = 1$ to N **do**
- 4: Sample $p_i \sim \mathcal{N}(0, I)$
- 5: Set $(q, p) \leftarrow (q_i^{(k-1)}, p_i)$; set $W_i \leftarrow 0$
- 6: **Half-kick** (potential & CD- q force):

$$p \leftarrow p - \frac{\varepsilon}{2} \left(\nabla_q U(q; \lambda_k) + \dot{\lambda}_k \nabla_q A_\phi(q, p; \lambda_k) \right)$$

- 7: **Drift** (kinetic & CD- p flow):

$$q \leftarrow q + \varepsilon \left(p + \dot{\lambda}_k \nabla_p A_\phi(q, p; \lambda_k) \right)$$

- 8: **Protocol work increment** (discrete $\partial_t H^{\text{CD}}$):

$$W_i \leftarrow W_i + \left(U(q; \lambda_{k+1}) - U(q; \lambda_k) \right) + \left(\dot{\lambda}_{k+1} A_\phi(q, p; \lambda_{k+1}) - \dot{\lambda}_k A_\phi(q, p; \lambda_k) \right)$$

- 9: **Half-kick** (with updated waypoint):

$$p \leftarrow p - \frac{\varepsilon}{2} \left(\nabla_q U(q; \lambda_{k+1}) + \dot{\lambda}_{k+1} \nabla_q A_\phi(q, p; \lambda_{k+1}) \right)$$

- 10: Update particle: $q_i^{(k)} \leftarrow q$
 - 11: Update weight: $w_i^{(k)} \leftarrow w_i^{(k-1)} \exp(-W_i)$
 - 12: **end for**
 - 13: Normalize weights: $w_i^{(k)} \leftarrow w_i^{(k)} / \sum_j w_j^{(k)}$
 - 14: Optionally resample if effective sample size is low
 - 15: **end for**
 - 16: **return** weighted population $\{(q_i^{(L)}, w_i^{(L)})\}_{i=1}^N$
-

180 **Numerical Integration** A key complication is that A is not separable into a term depending on p
181 and a term depending on q . This means that the Velocity Verlet integrator is not symplectic.

182 **Choice of step size** As in standard HMC, the choice of ε is a trade-off between the accuracy of the
183 dynamics and the efficiency of the algorithm. We propose to tune step size by monitoring the error in
184 H ; in the limit of small step size and perfect A , H should be preserved exactly.

185 consider fisher score

5 Evaluation

The goal of CHMC is to reduce the wall-clock time required to sample from challenging distributions. We consider the regime where the gradient of the log density is costly, so that our central focus is on the number of calls to this gradient function required for an *effective sample size* of 100 samples.

We compare to Sequential Monte Carlo, as well as the results reported in NETS TODO

We use the models proposed in the NETS paper for evaluation (todo: get michael or eric to share them).

6 Conclusion

A Counterdiabatic HMC as an SMC sampler

Given an initial distribution $\mu \in \mathcal{P}(\mathbf{X})$ and a sequence of Markov kernels $M_t \in \mathcal{M}(\mathbf{X})$ for $t = 1, \dots, T$, the *proposal* (or reference) path measure $\mathbb{Q}$ is

$$\mathbb{Q}(\mathrm{d}x_{0:T}) = \mu(\mathrm{d}x_0) \prod_{t=1}^T M_t(x_{t-1}, \mathrm{d}x_t). \quad (3)$$

Importance sampling on path space (Feynman–Kac). Let $G_0 : \mathbf{X} \rightarrow (0, \infty)$ and $G_t : \mathbf{X} \times \mathbf{X} \rightarrow (0, \infty)$ for $t \geq 1$ be strictly positive *potentials*. The *target* path measure $\mathbb{P}$ is defined by reweighting $\mathbb{Q}$:

$$\mathbb{P}(\mathrm{d}x_{0:T}) = \frac{1}{Z} G_0(x_0) \left[\prod_{t=1}^T G_t(x_{t-1}, x_t) \right] \mathbb{Q}(\mathrm{d}x_{0:T}), \quad (4)$$

where the normalizing constant is $Z = \mathbb{E}_{\mathbb{Q}} \left[G_0(X_0) \prod_{t=1}^T G_t(X_{t-1}, X_t) \right]$. The resulting unnormalized *marginal* measures $\gamma_t \in \mathcal{S}(\mathbf{X})$ and their normalized counterparts $\eta_t \in \mathcal{P}(\mathbf{X})$ are

$$\gamma_t(\varphi) = \mathbb{E}_{\mathbb{Q}} \left[\varphi(X_t) G_0(X_0) \prod_{s=1}^t G_s(X_{s-1}, X_s) \right], \quad Z_t := \gamma_t(1), \quad (5)$$

$$\eta_t(\varphi) = \gamma_t(\varphi) / Z_t, \quad t = 0, \dots, T. \quad (6)$$

Eqs. (3)–(5) constitute the standard Feynman–Kac framework: importance sampling is performed on the *path* measure, while particle approximations target the time marginals $(\eta_t)_{t=0:T}$.

One can then sample from $\mathbb{P}$ by initializing a population of particles from π_0 and then propagating them forward in time using the kernel M_t , keeping track of the weights at each step and optionally resampling to mitigate weight degeneracy.

Sequential Monte Carlo samplers [doucet2001sequential] show how the general framework of Sequential Monte Carlo as summarized above can be applied to the problem of sampling from a distribution π on $\mathbb{R}^d$, by artificially constructing a path of distributions π_t with π_0 a simple distribution and $\pi_T = \pi$ the target distribution.

The central idea is that a forward path can be constructed by choosing a time dependent Markov kernel M_t

$\mathbb{Q}(\mathbf{M}, \pi_0)$

and that this can be renormalized to sample from a

$\mathbb{Q}(\mathbf{L}, \pi_T)$

This requires $G = \dots$

SMC with physical dynamics: nonequilibrium fluctuation theorems In the case where M and L are given by forward time and backward time Hamiltonian dynamics (or overdamped or underdamped Langevin dynamics), the ratio of $M(x_t|x_{t-1})$ to $L(x_{t-1}|x_t)$ can be calculated using the non-equilibrium fluctuation theorem derived by [7] and extended to more generality [todo].

221 First, recall that

$$\frac{dH}{dt} = \frac{\partial H}{\partial t} + \nabla H \cdot \frac{d\mathbf{x}}{dt} := dW + dQ$$

222 In physics terms, dW and dQ are the work done by the system and the heat absorbed by the system,
 223 respectively, and mathematically, should be viewed as one-forms that are not exact, i.e. are not the
 224 differential of a function. By contrast, their sum dH is an exact one-form, namely the differential of
 225 the Hamiltonian.

226 In particular, this theorem states that

$$\frac{\pi(z_{t-1})k(z_t|z_{t-1})}{\pi(z_t)k(z_{t-1}|z_t)} = \exp^{W-\Delta F}$$

227 where W is the integral of $dw = \frac{\partial H}{\partial t}$ over the trajectory, and $\Delta F = -\log \frac{Z_t}{Z_{t-1}}$, for $Z_t :=$
 228 $\int \pi_{H(t)}(z)dz$.

229 The consequence is that we can compute

230 TODO: define k

$$\frac{k(z_t|z_{t-1})}{k(z_{t-1}|z_t)} = e^W \frac{Z_t}{Z_{t-1}} \frac{Z_{t-1}}{Z_t} e^{-\Delta H} = e^{W-\Delta H} = e^{-Q}$$

231 Since Hamiltonian dynamics are measure-preserving, i.e. $Q = 0$, even when numerically simulated
 232 with a leapfrog integrator, we see immediately that when using an unadjusted HMC kernel in the
 233 SMC sampling algorithm, the correct weight is given by $e^{-\Delta H}$, in agreement with Appendix B of
 234 [8].

235 The more interesting case concerns a kernel k that arises from a counterdiabatic Hamiltonian H_c . In
 236 this case, [35] derives an analogous theorem, but in terms of a different work, which we refer to as
 237 W_c , obtained by integrating $dw_c = \text{todo}$ over the trajectory, where

$$dw_c = \left(\frac{\partial H_c}{\partial t} - \nabla H \cdot v + \nabla \cdot v \right) dt$$

238 By definition of

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [NA].
- [NA] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for NA).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will be asked to also include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While "[Yes]" is generally preferable to "[No]", it is perfectly acceptable to answer "[No]" provided a proper justification is given (e.g., "error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering "[No]" or "[NA]" is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist",
- Keep the checklist subsection headings, questions/answers and guidelines below.
- Do not modify the questions and only use the provided macros for your answers.

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the paper does not include experiments.

- If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so “No” is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyper-parameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the paper does not include experiments.
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
- If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.

- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that there is no societal impact of the work performed.
- If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer NA means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer NA means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy (<https://neurips.cc/Conferences/2025/LLM>) for what should or should not be described.

References

- [1] Michael S Albergo and Eric Vanden-Eijnden. “Nets: A non-equilibrium transport sampler”. In: *arXiv preprint arXiv:2410.02711* (2024).
- [2] Michael Arbel, Alex Matthews, and Arnaud Doucet. “Annealed flow transport monte carlo”. In: *International Conference on Machine Learning*. PMLR. 2021, pp. 318–330.
- [3] Espen Bernton et al. “Schrodinger Bridge Samplers”. In: *arXiv preprint arXiv:1912.13170* (2019).
- [4] Michael Victor Berry. “Transitionless quantum driving”. In: *Journal of Physics A: Mathematical and Theoretical* 42.36 (2009), p. 365303.
- [5] Michael Betancourt. “A conceptual introduction to Hamiltonian Monte Carlo”. In: *arXiv preprint arXiv:1701.02434* (2017).

- 596 [6] Siddhartha Chib and Edward Greenberg. “Understanding the metropolis-hastings algorithm”.
597 In: *The american statistician* 49.4 (1995), pp. 327–335.
- 598 [7] Gavin E Crooks. “Nonequilibrium measurements of free energy differences for microscopically
599 reversible Markovian systems”. In: *Journal of statistical physics* 90.5 (1998), pp. 1481–1487.
- 600 [8] Chenguang Dai et al. “An invitation to sequential Monte Carlo samplers”. In: *Journal of the
601 American Statistical Association* 117.539 (2022), pp. 1587–1600.
- 602 [9] Pierre Del Moral, Arnaud Doucet, and Ajay Jasra. “Sequential monte carlo samplers”. In:
603 *Journal of the Royal Statistical Society Series B: Statistical Methodology* 68.3 (2006), pp. 411–
604 436.
- 605 [10] Mustafa Demirplak and Stuart A Rice. “Adiabatic population transfer with control fields”. In:
606 *The Journal of Physical Chemistry A* 107.46 (2003), pp. 9937–9945.
- 607 [11] Arnaud Doucet, Nando De Freitas, and Neil Gordon. “An introduction to sequential Monte
608 Carlo methods”. In: *Sequential Monte Carlo methods in practice* (2001), pp. 3–14.
- 609 [12] Mingzhou Fan et al. “Path-guided particle-based sampling”. In: *arXiv preprint
610 arXiv:2412.03312* (2024).
- 611 [13] Wendell H Fleming and Raymond W Rishel. *Deterministic and stochastic optimal control*.
612 Vol. 1. Springer Science & Business Media, 2012.
- 613 [14] Dani Gamerman and Hedibert F Lopes. *Markov chain Monte Carlo: stochastic simulation for
614 Bayesian inference*. Chapman and Hall/CRC, 2006.
- 615 [15] Andrew Gelman, Daniel Lee, and Jiqiang Guo. “Stan: A probabilistic programming language
616 for Bayesian inference and optimization”. In: *Journal of Educational and Behavioral Statistics*
617 40.5 (2015), pp. 530–543.
- 618 [16] David Guery-Odelin et al. “Driving rapidly while remaining in control: classical shortcuts
619 from Hamiltonian to stochastic dynamics”. In: *Reports on Progress in Physics* 86.3 (2023),
620 p. 035902.
- 621 [17] David Guery-Odelin et al. “Shortcuts to adiabaticity: Concepts, methods, and applications”.
622 In: *Reviews of Modern Physics* 91.4 (2019), p. 045001.
- 623 [18] Jeremy Heng et al. “Controlled sequential monte carlo”. In: *The Annals of Statistics* 48.5
624 (2020), pp. 2904–2929.
- 625 [19] Matthew D Hoffman, Andrew Gelman, et al. “The No-U-Turn sampler: adaptively setting path
626 lengths in Hamiltonian Monte Carlo.” In: *J. Mach. Learn. Res.* 15.1 (2014), pp. 1593–1623.
- 627 [20] Christopher Jarzynski. “Nonequilibrium equality for free energy differences”. In: *Physical
628 Review Letters* 78.14 (1997), p. 2690.
- 629 [21] Michael Kolodrubetz et al. “Geometry and non-adiabatic response in quantum and classical
630 systems”. In: *Physics Reports* 697 (2017), pp. 1–87.
- 631 [22] Ignacio A Martinez et al. “Engineered swift equilibration of a Brownian particle”. In: *Nature
632 physics* 12.9 (2016), pp. 843–846.
- 633 [23] Radford M Neal. “Annealed importance sampling”. In: *Statistics and computing* 11 (2001),
634 pp. 125–139.
- 635 [24] Ayoti Patra and Christopher Jarzynski. “Shortcuts to adiabaticity using flow fields”. In: *New
636 Journal of Physics* 19.12 (2017), p. 125009.
- 637 [25] Gabriel Peyre, Marco Cuturi, et al. “Computational optimal transport: With applications to
638 data science”. In: *Foundations and Trends® in Machine Learning* 11.5-6 (2019), pp. 355–607.
- 639 [26] Dries Sels and Anatoli Polkovnikov. “Minimizing irreversible losses in quantum systems by
640 local counterdiabatic driving”. In: *Proceedings of the National Academy of Sciences* 114.20
641 (2017), E3909–E3916.
- 642 [27] David A Sivak, John D Chodera, and Gavin E Crooks. “Using Nonequilibrium Fluctuation
643 Theorems to Understand and Correct Errors in Equilibrium and Nonequilibrium Simulations
644 of Discrete Langevin Dynamics”. In: *Physical Review X* 3.1 (2013), p. 011007.
- 645 [28] Jingtong Sun et al. “Dynamical measure transport and neural PDE solvers for sampling”. In:
646 *arXiv preprint arXiv:2407.07873* (2024).
- 647 [29] Yifeng Tian, Nishant Panda, and Yen Ting Lin. “Liouville flow importance sampler”. In: *arXiv
648 preprint arXiv:2405.06672* (2024).
- 649 [30] Mark E Tuckerman. *Statistical mechanics: theory and molecular simulation*. Oxford university
650 press, 2023.

- 651 [31] Suriyanarayanan Vaikuntanathan and Christopher Jarzynski. “Escorted free energy simula-
652 tions”. In: *The Journal of chemical physics* 134.5 (2011).
- 653 [32] Francisco Vargas et al. “Transport meets variational inference: Controlled monte carlo diffu-
654 sions”. In: *arXiv preprint arXiv:2307.01050* (2023).
- 655 [33] Udo Von Toussaint. “Bayesian inference in physics”. In: *Reviews of Modern Physics* 83.3
656 (2011), pp. 943–999.
- 657 [34] Qinsheng Zhang and Yongxin Chen. “Path integral sampler: a stochastic control approach for
658 sampling”. In: *arXiv preprint arXiv:2111.15141* (2021).
- 659 [35] Adrienne Zhong, Ben Kuznets-Speck, and Michael R DeWeese. “Time-asymmetric fluctuation
660 theorem and efficient free-energy estimation”. In: *Physical Review E* 110.3 (2024), p. 034121.